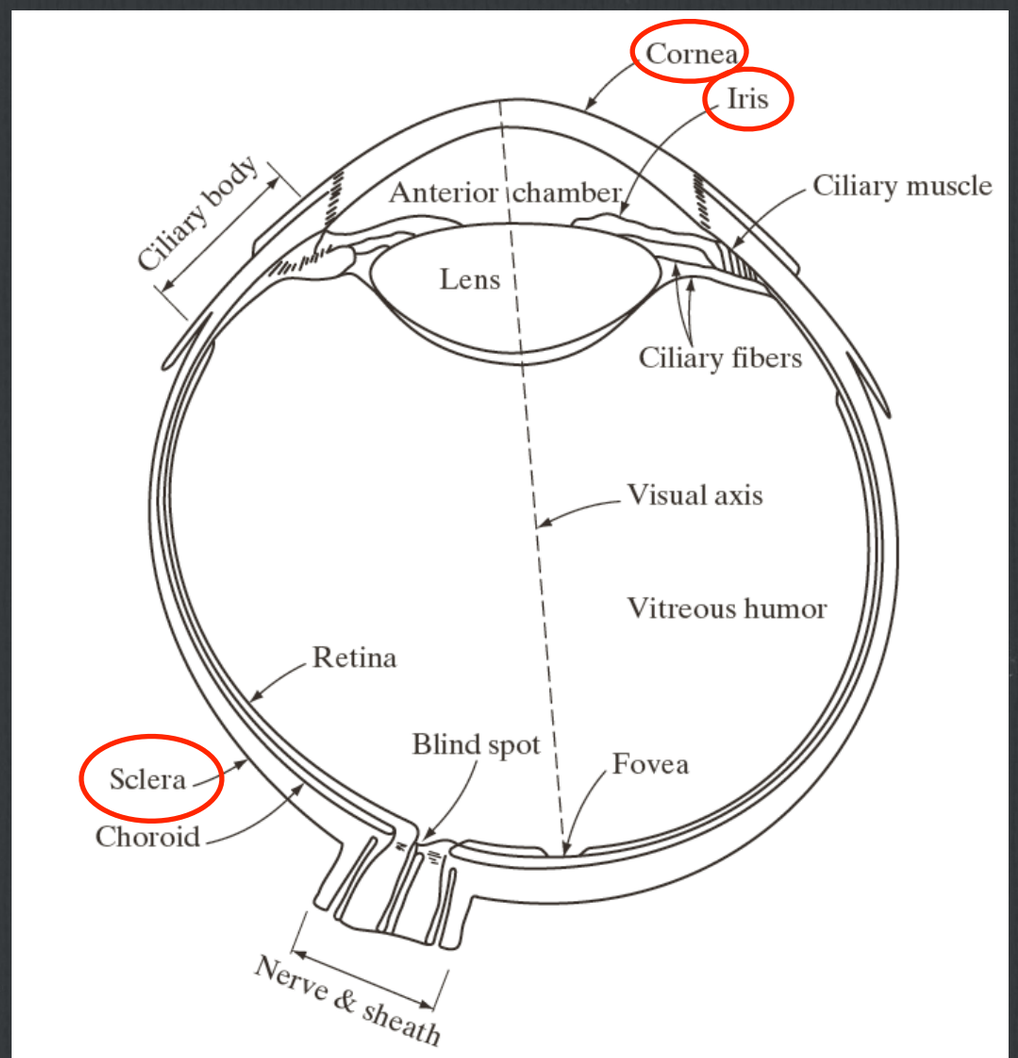


Digital Image Processing

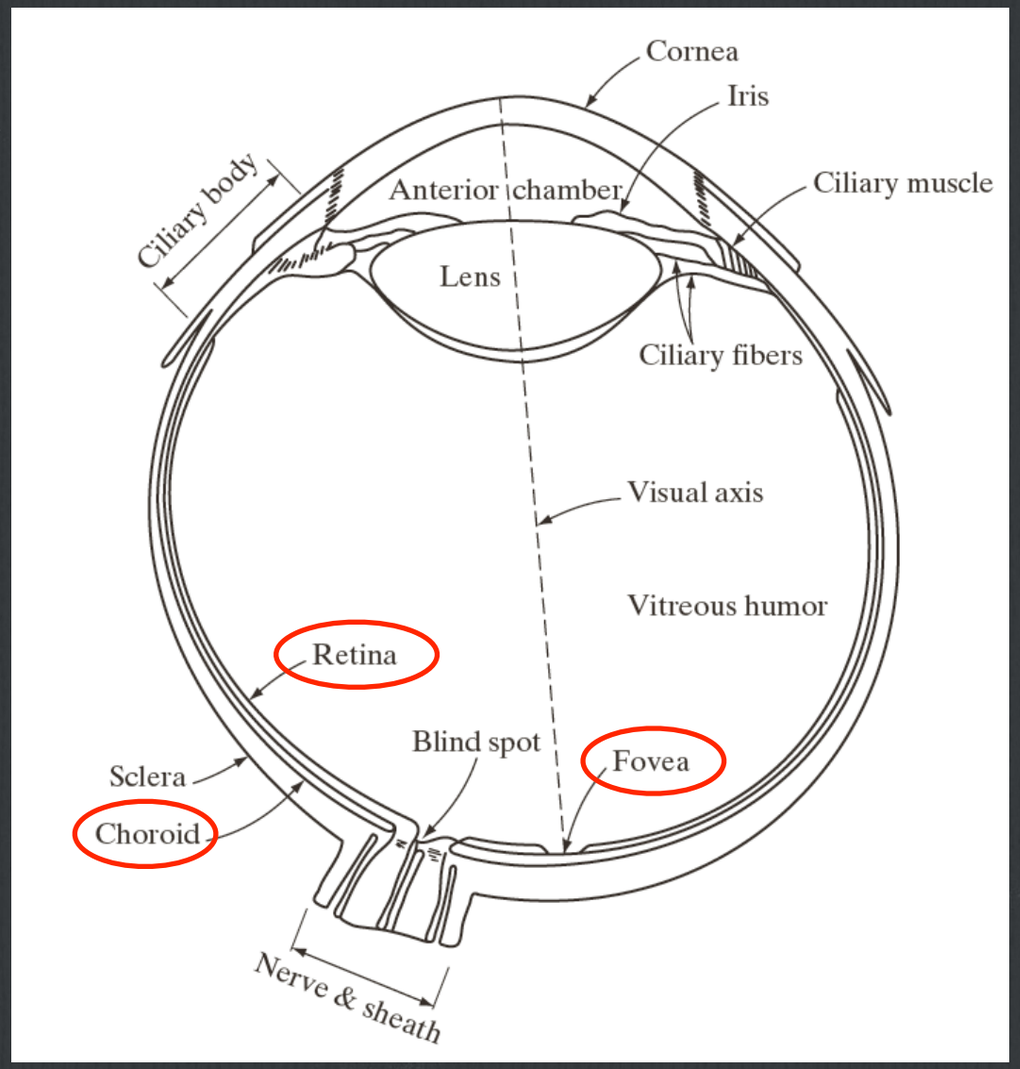
Chapter 2 Digital Image Fundamentals

Elements of Visual Perception

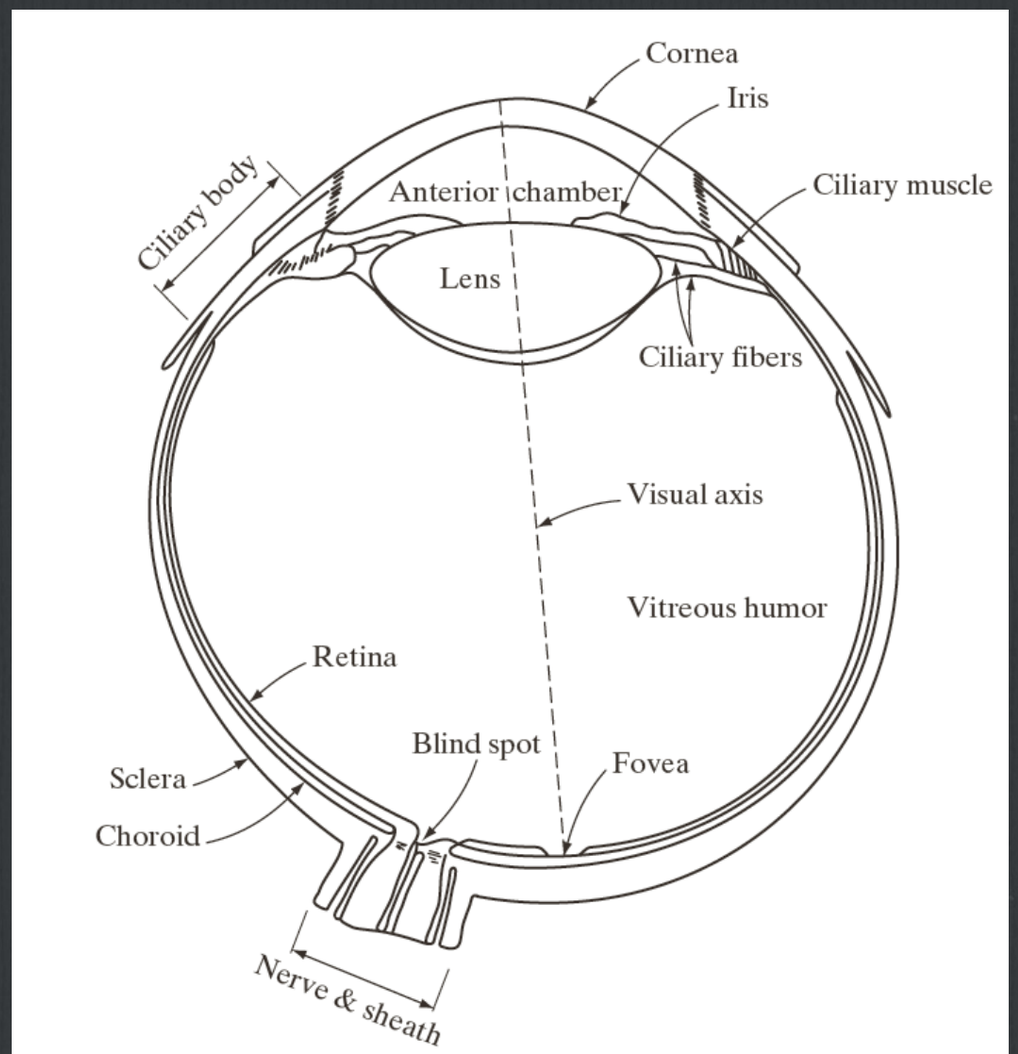
Structure of the Human Eye



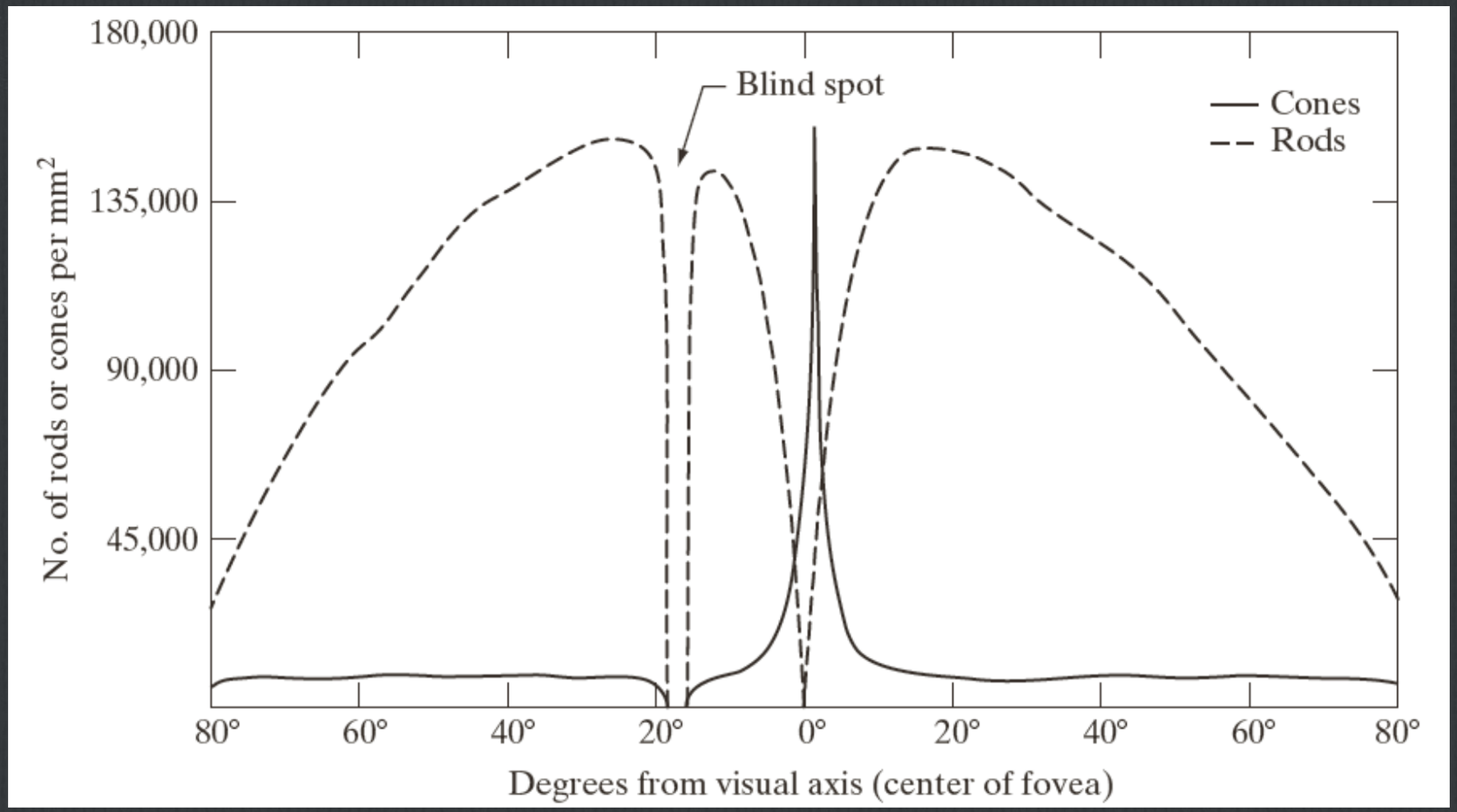
- Average diameter ~20 mm.
- Iris (ม่านตา) acts as a diaphragm to control the amount of light entering the eye.
  - Eye color: melanin (pigment) in iris
- Light entering the cornea is focused to the retina surface by Lens
- 3 membranes enclose the eye
  - Outer cover:
    - cornea (กระจกตา) - a tough, transparent tissue that cover the anterior surface of the eye.
    - sclera (นัยน์ตาขาว) - an opaque membrane that encloses the reminder of the optic globe.



- ☐ **Choroid**
  - ☐ lies directly below the sclera
  - ☐ Contains a network of blood vessels that serve as the major source of nutrition to the eye
- ☐ **Retina (จอประสาทตา)**
  - ☐ Lines the inside of the wall's entire posterior portion.
  - ☐ When the eye is properly focused, light from an object outside the eye is imaged on the retina.
  - ☐ 2 classes of receptors
    - ☐ Cones - in each eye ~6-7 milloin
      - ☐ Located primarily in the central portion of the retina, fovea
      - ☐ Highly sensitive to color
      - ☐ Cone vision is called photopic or bright-light vision.



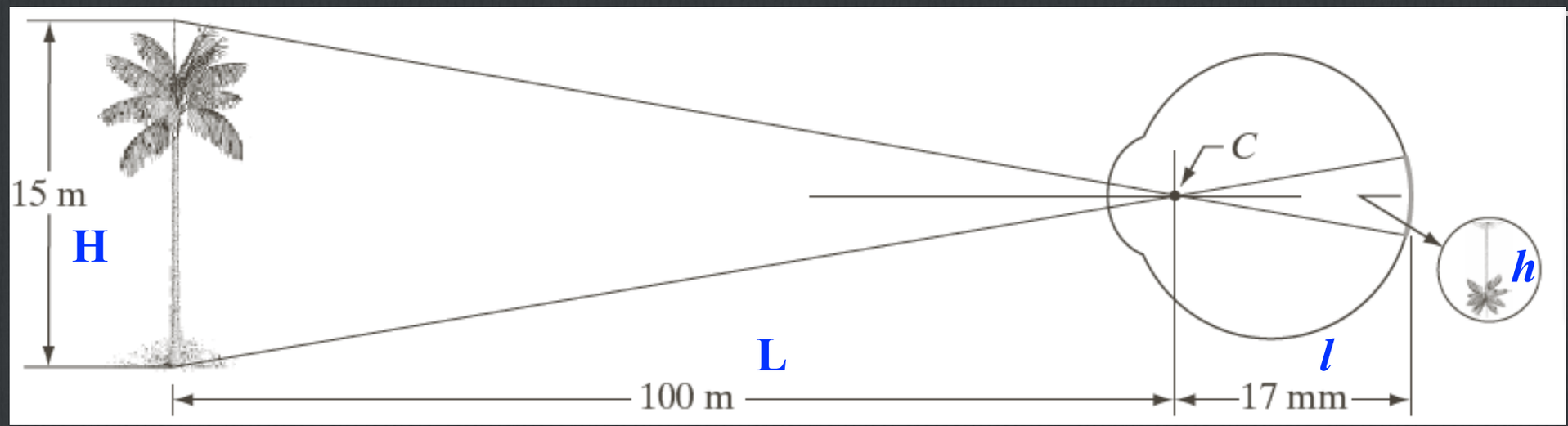
- Retina (จอประสาทตา)
  - 2 classes of receptors
    - Rods - ~75-150 milloin
      - Serve to give a general, overall picture of the field of view.
      - Not involved in color vision.
      - Sensitive to low levels of illumination
      - Ex. Objects that appear brightly color in daylight when seen by moonlight appear as colorless forms because only the rods are stimulated.
      - This phenomenon is known as scotopic or dim-light vision.



Distribution of rods and cones in the retina.

- The distribution of rods and cones is radially symmetric wrt the **fovea** (central portion of the retina), except at the **blind spot** which includes no receptors.
- Fovea area in the retina is circular with 1.5 mm in diameter where most of the cones are concentrated with 150,000 cones/mm^2.

Image Formation in the Eye



Graphical representation of the eye looking at a palm tree. Point C is the optical center of the lens

- Focal length
  - distance between centre of the lens and the retina
  - Varies from 14 mm to 17 mm
  - H/L = h/l

- Perception takes place by the relative excitation of light receptors, which transform radiant energy into electrical impulses that are ultimately decoded by the brain.

Image Acquisition

A simple image formation model

- Image: a 2-D light-intensity function f(x,y)
- The value of f at (x,y) → the intensity (brightness) of the image at that point
- $0 < f(x,y) < \infty$

- Consider the monochrome case, e.g., black and white images

- Represent the spectral intensity distribution of the image by a continuous function f(x,y), i.e., for fixed value of (x,y), f(x,y) is proportional to the grey level of the image at that point.

- Of course,

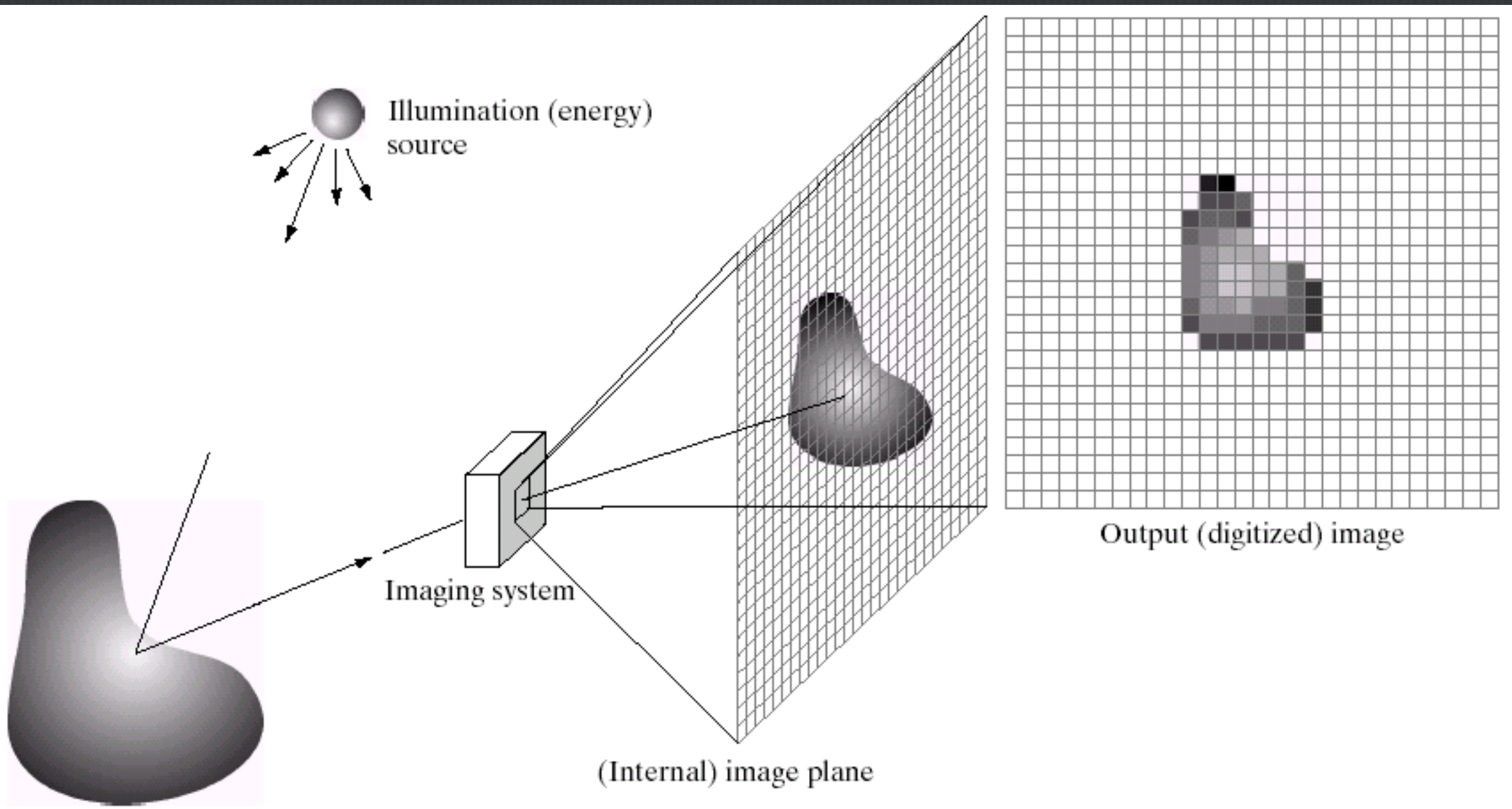
$$\text{(black) } 0 < f(x,y) < f_{max} \text{ (white)}$$

- Intermediate values between 0 and f_{max} are called shades of gray varying from black to white.

- f(x,y) may be characterized by 2 components:
  - Amount of source illumination incident on the scene being viewed: illumination, i(x,y)
  - Amount of illumination reflected by the objects in the scene: reflectance, r(x,y)

$$f(x,y) = i(x,y)r(x,y)$$

- $0 < i(x,y) < \infty$, $0 < r(x,y) < 1$

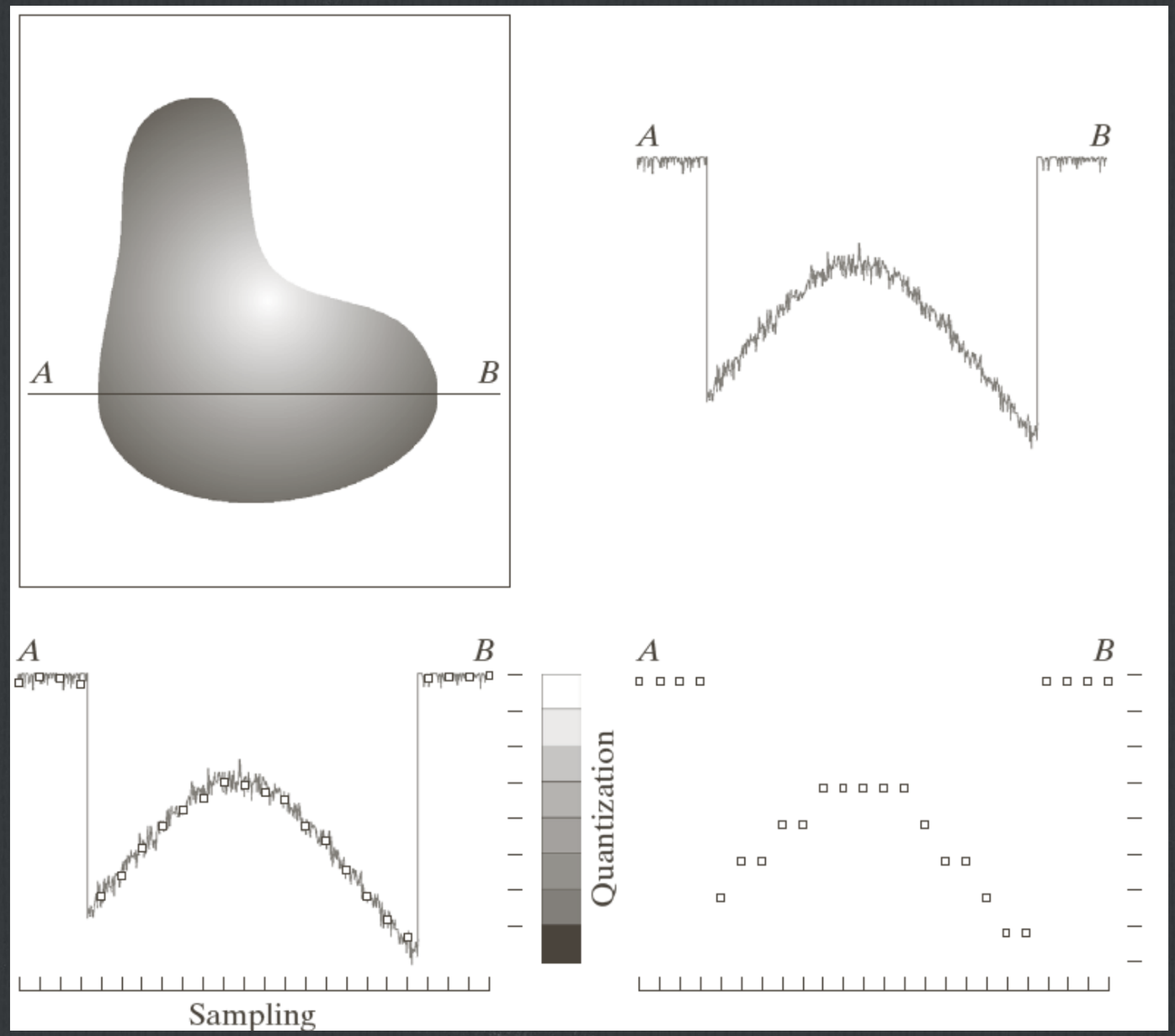


a
b c d e

FIGURE 2.15 An example of the digital image acquisition process. (a) Energy ("illumination") source. (b) An element of a scene. (c) Imaging system. (d) Projection of the scene onto the image plane. (e) Digitized image.

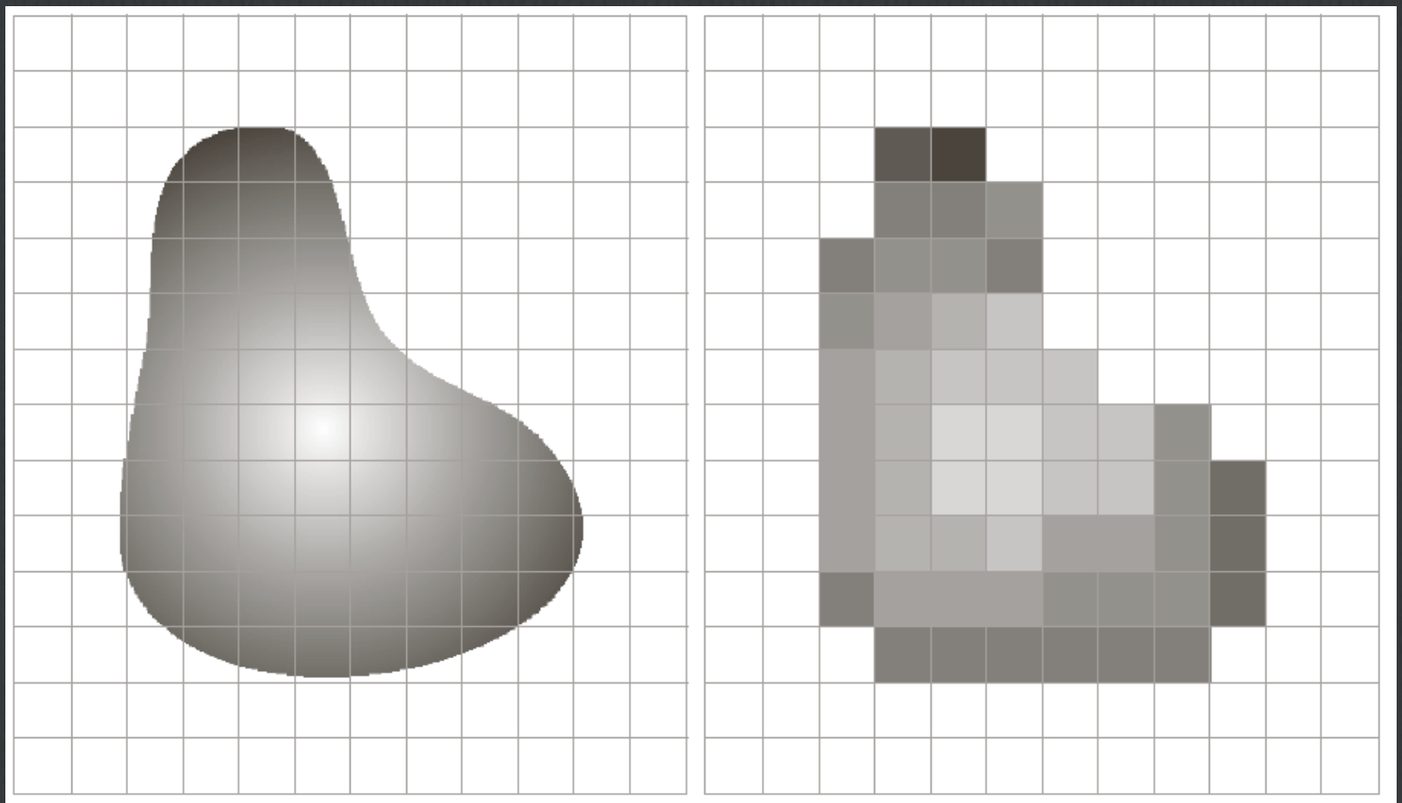
Basic Concepts in Sampling and Quantization

Basic Concepts in Sampling and Quantization



a b
c d

FIGURE 2.16 Generating a digital image. (a) Continuous image. (b) A scan line from *A* to *B* in the continuous image, used to illustrate the concepts of sampling and quantization. (c) Sampling and quantization. (d) Digital scan line.

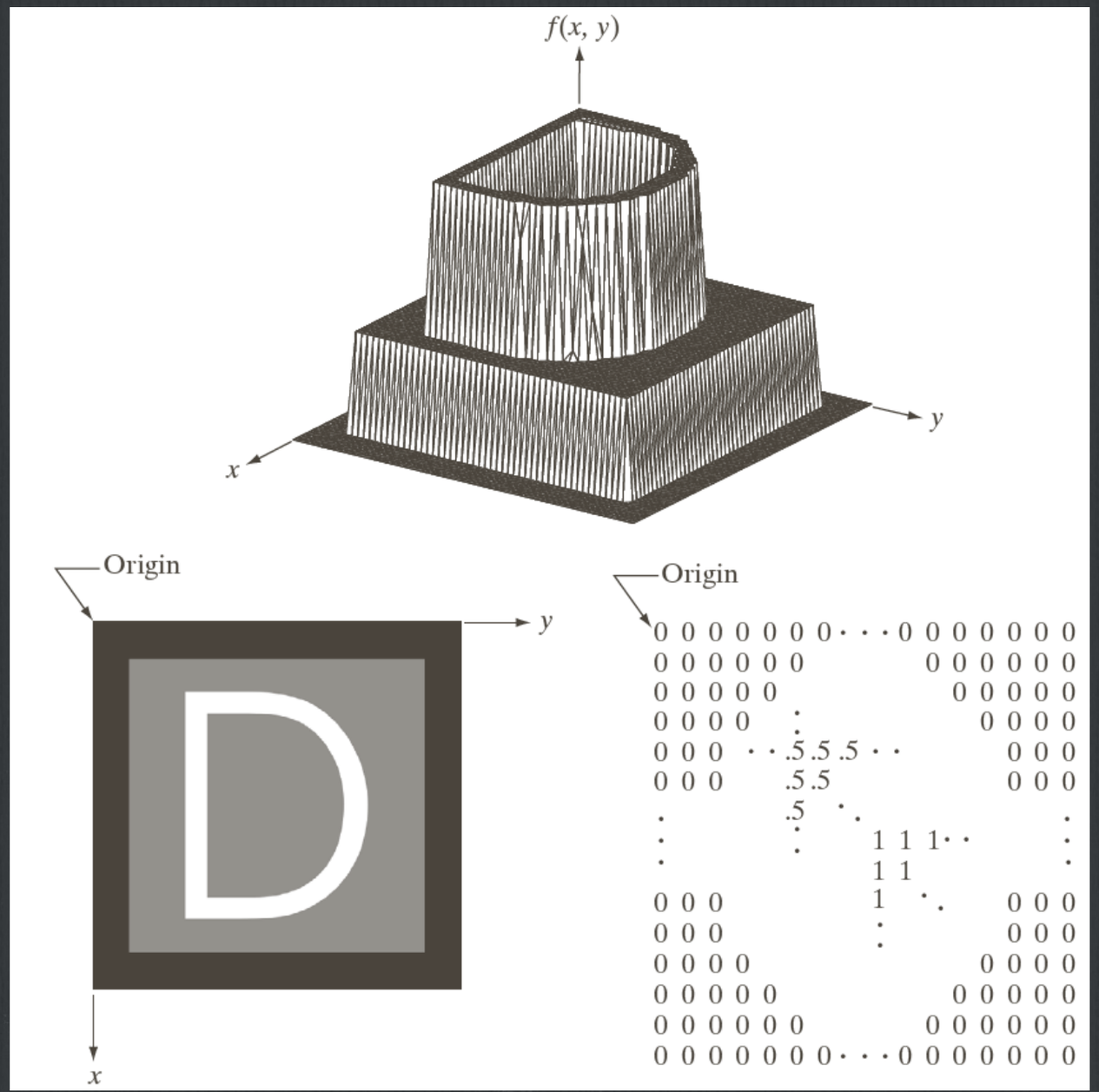


a b

FIGURE 2.17 (a) Continuous image projected onto a sensor array. (b) Result of image sampling and quantization.

- The spatial and amplitude digitization of f(x,y) is called:
  - **Image sampling** when it refers to <u>spatial coordinates (x,y)</u>
  - **Gray-level quantization** when it refers to the <u>amplitude</u>

Representing Digital Image



a
b c

FIGURE 2.18
(a) Image plotted as a surface.
(b) Image displayed as a visual intensity array.
(c) Image shown as a 2-D numerical array (0, .5, and 1 represent black, gray, and white, respectively).

$$f(x,y)=\begin{bmatrix} f(0,0) & f(0,1) & \dots & f(0,N-1) \\ f(1,0) & f(1,1) & \dots & f(1,N-1) \\ \vdots & \vdots & & \vdots \\ f(M-1,0) & f(M-1,1) & \dots & f(M-1,N-1) \end{bmatrix}$$

Digital image

Image elements (pixels)

- Z: set of real integers
- R: set of real numbers
- Sampling process: partitioning the xy-plane into a grid
  - The coordinate of the center of each grid is a pair of elements from the Cartesian product ZxZ (Z^2)
  - Z^2 is the set of all ordered pairs of elements (z_i,z_j) with z_i and z_j being integers from Z.

- f(x,y) is a digital image if:
  - (x,y) are integers from Z^2 and
  - f is a function that assigns a gray-level value (from R) to each distinct pair of coordinates (x,y) [quantization]
- Gray levels are usually integers
  - then Z replaces R

- The digitization process requires decisions about:
  - values for N,M (where N x M: the image array)
  - the number (L) of discrete gray levels allowed for each pixel
- Usually, in DIP these quantities are integer powers of two:

$$N = 2^n \qquad M = 2^m \qquad L = 2^k$$

- Another assumption is that the discrete levels are equally spaced between 0 and L-1 in the gray scale.

- **Dynamic range**: the ratio of the maximum measurable intensity to the minimum detectable intensity level in the system
- Upper limit - **saturation**
- Lower limit - **noise**
- Basically, dynamic range establishes the lowest and highest intensity levels that a system can represent, and can have
- **Contrast** - the difference in intensity between the highest and lowest intensity levels in an image

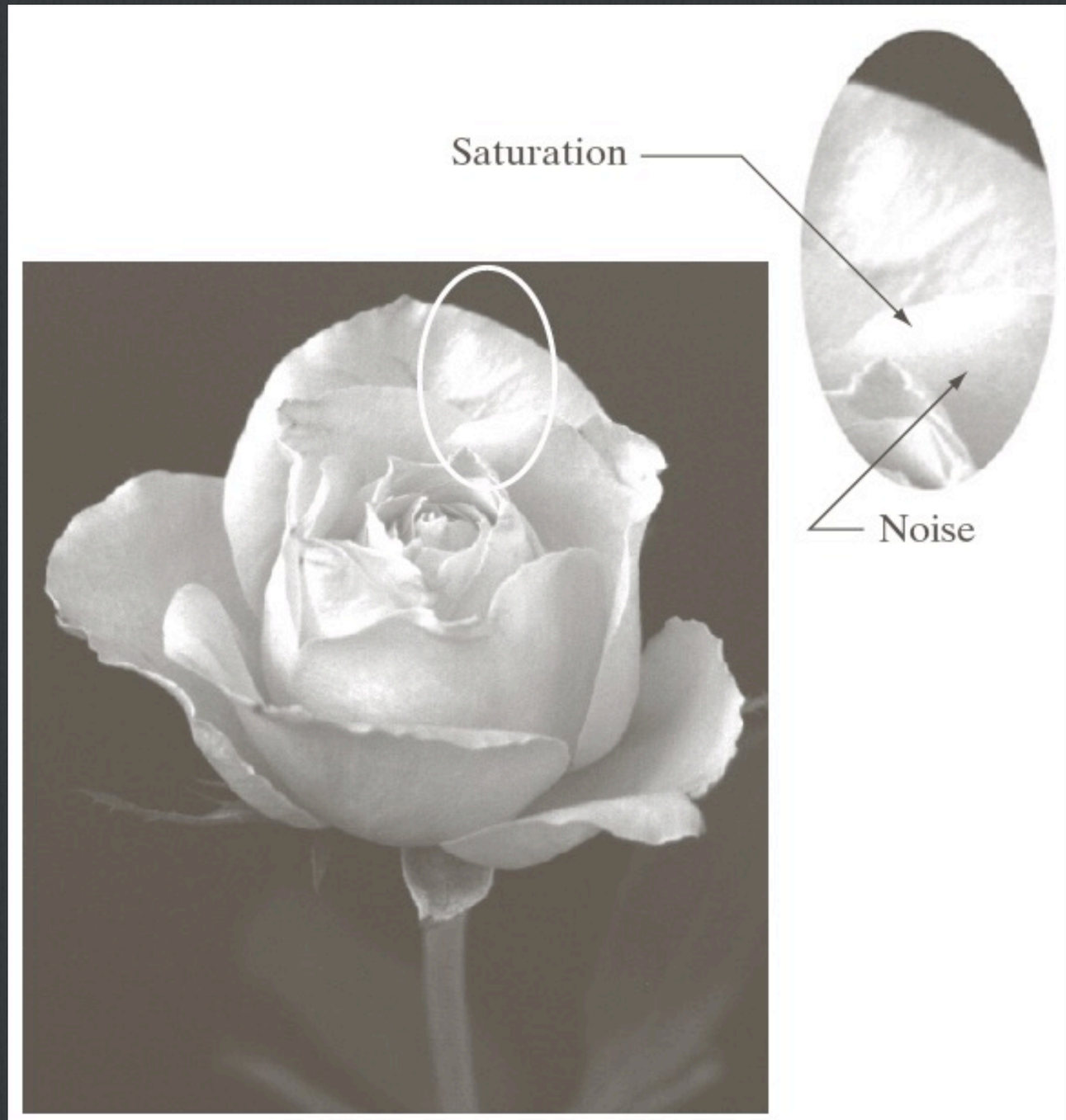


FIGURE 2.19 An image exhibiting saturation and noise. Saturation is the highest value beyond which all intensity levels are clipped (note how the entire saturated area has a high, *constant* intensity level). Noise in this case appears as a grainy texture pattern. Noise, especially in the darker regions of an image (e.g., the stem of the rose) masks the lowest detectable true intensity level.

- If b is the number of bits required to store a digitized image then:

$$b = NxMxk \text{ (if } M=N\text{, then } b=N^2k)$$

TABLE 2.1
Number of storage bits for various values of N and k.

N/k	1 ($L = 2$)	2 ($L = 4$)	3 ($L = 8$)	4 ($L = 16$)	5 ($L = 32$)	6 ($L = 64$)	7 ($L = 128$)	8 ($L = 256$)
32	1,024	2,048	3,072	4,096	5,120	6,144	7,168	8,192
64	4,096	8,192	12,288	16,384	20,480	24,576	28,672	32,768
128	16,384	32,768	49,152	65,536	81,920	98,304	114,688	131,072
256	65,536	131,072	196,608	262,144	327,680	393,216	458,752	524,288
512	262,144	524,288	786,432	1,048,576	1,310,720	1,572,864	1,835,008	2,097,152
1024	1,048,576	2,097,152	3,145,728	4,194,304	5,242,880	6,291,456	7,340,032	8,388,608
2048	4,194,304	8,388,608	12,582,912	16,777,216	20,971,520	25,165,824	29,369,128	33,554,432
4096	16,777,216	33,554,432	50,331,648	67,108,864	83,886,080	100,663,296	117,440,512	134,217,728
8192	67,108,864	134,217,728	201,326,592	268,435,456	335,544,320	402,653,184	469,762,048	536,870,912

Spatial Resolution

- **Spatial resolution:**
  - A measure of the smallest discernible detail in an image.
  - Number of samples per unit length or area.
  - **Dot (pixels) per unit distance / Dot Per Inch (DPI)**: specifies the size of an individual pixel, used commonly in the printing and publishing industry.
  - To be meaningful, measures of spatial resolution must be stated with respect to spatial units.
  - Image size by itself does not tell the complete story.



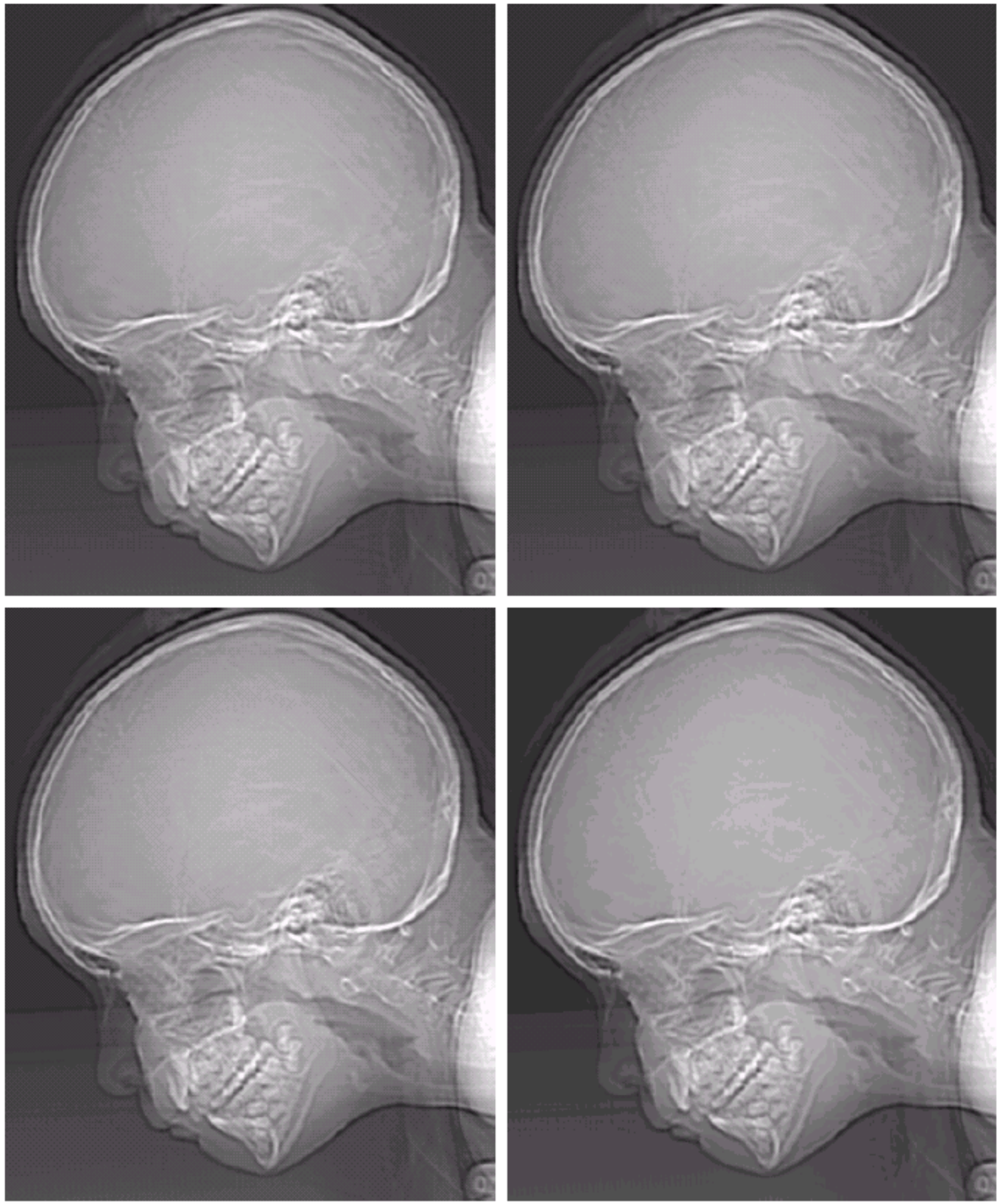
a b
c d

FIGURE 2.20 Typical effects of reducing spatial resolution. Images shown at: (a) 1250 dpi, (b) 300 dpi, (c) 150 dpi, and (d) 72 dpi. The thin black borders were added for clarity. They are not part of the data.

If the resolution is decreased too much, the checkerboard effect can occur.

Intensity Resolution

- **Intensity resolution:**
  - Similarly refers to the smallest discernible change in intensity level.
    - Number of bits per pixel
    - Usually 8 bits
    - Color image has 3 image planes to yield 8x3 = 24 bits/pixel
    - Too few levels may cause false contour
    - If the gray scale is not enough, the smooth area will be affected.
    - False contouring can occur on the smooth area which has fine gray scales.

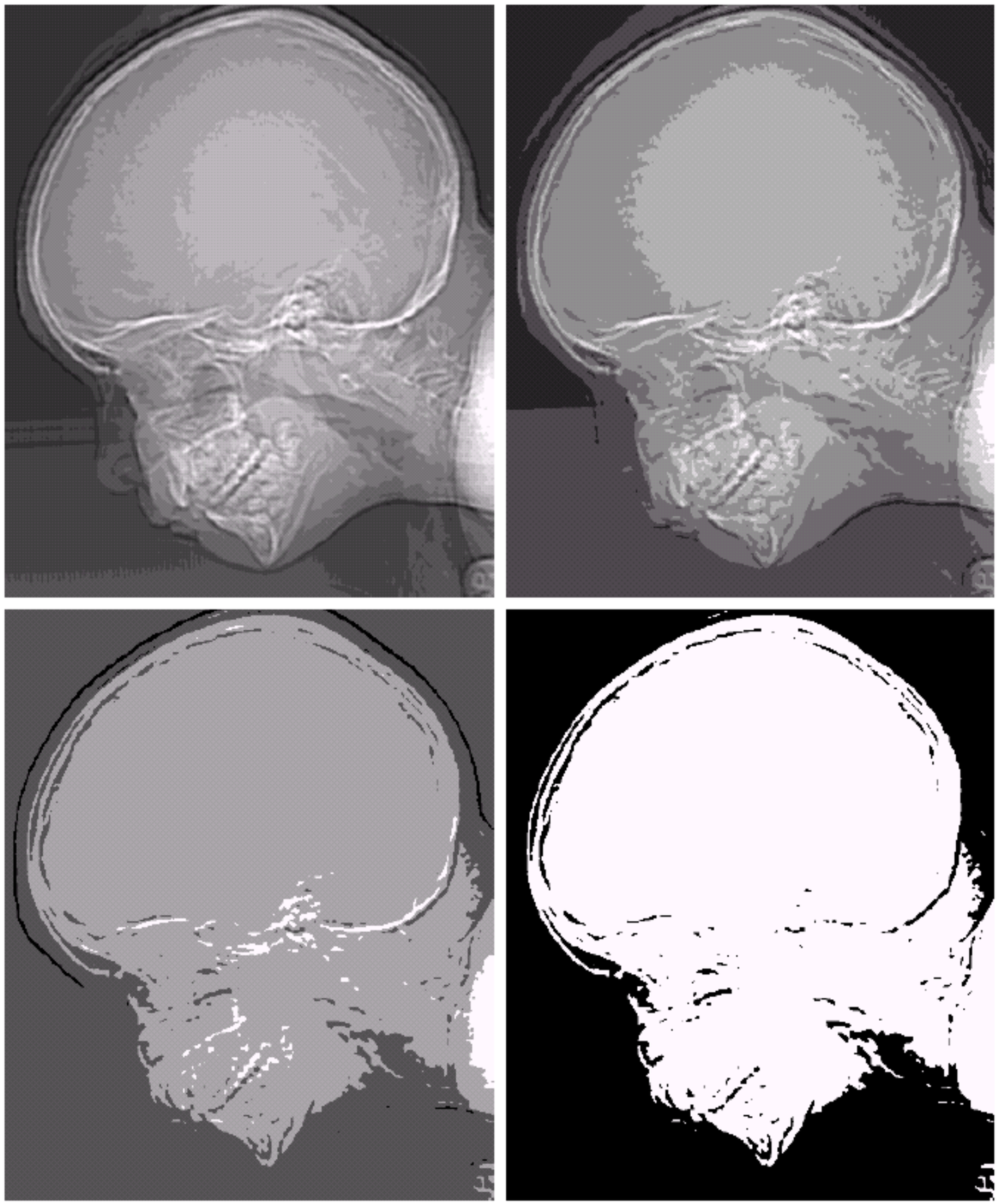


a b
c d

FIGURE 2.21
(a) 452 × 374, 256-level image. (b)–(d) Image displayed in 128, 64, and 32 gray levels, while keeping the spatial resolution constant.

e f
g h

FIGURE 2.21
(Continued)
(e)–(h) Image displayed in 16, 8, 4, and 2 gray levels. (Original courtesy of Dr. David R. Pickens, Department of Radiology & Radiological Sciences, Vanderbilt University Medical Center.)



Nonuniform Sampling

- For a fixed value of spatial resolution, the appearance of the image can be improved by using adaptive sampling rates.
  - Fine sampling ⇒ required in the neighborhood of sharp gray-level transitions.
  - Coarse sampling ⇒ utilized in relatively smooth regions.



a b c

FIGURE 2.22 (a) Image with a low level of detail. (b) Image with a medium level of detail. (c) Image with a relatively large amount of detail. (Image (b) courtesy of the Massachusetts Institute of Technology.)

- An an image with a face superimposed on a uniform background.
  - background ⇒ little detailed information ⇒ coarse sampling is enough.
  - face ⇒ more detail ⇒ fine sampling.
- If we can use adaptive sampling, the quality of the image is improved.
- Moreover, we should care more around the boundary of the object sharp gray-level transmission from object to background.

Some basic relationships between pixels

Neighbors of a pixel

- A pixel p at coordinates (x,y) has
- $N_4(p)$, the 4-neighbors of p: 4 horizontal and vertical neighbors

  (x+1, y), (x-1, y), (x, y+1), (x, y-1)
- $N_D(p)$, the 4-diagonal of p:

  (x+1, y+1), (x+1, y-1), (x-1, y+1), (x-1, y-1)
- $N_8(p)$, the 8-neighbors of p: a combination of $N_4(p)$ and $N_D(p)$

Adjacency, Connectivity, Regions, and Boundaries

- V : set of intensity values and used to defined adjacency.
  - In a **binary image**: V = {1} if we are referring to adjacency of pixels with value 1.
  - In a **gray-scale image**: the idea is the same, but set V typically contains more elements, e.g. V={120, 121, 122,…, 140}

☐ 3 types of adjacency:

A. **4-adjacency**: two pixel p and q with values from V are 4-adjacent if q is in the set $N_4(p)$.

B. **8-adjacency**: two pixel p and q with values from V are 8-adjacent if q is in the set $N_8(p)$.

C. **m-adjacency (mixed adjacency)**: two pixel p and q with values from V are m-adjacent if

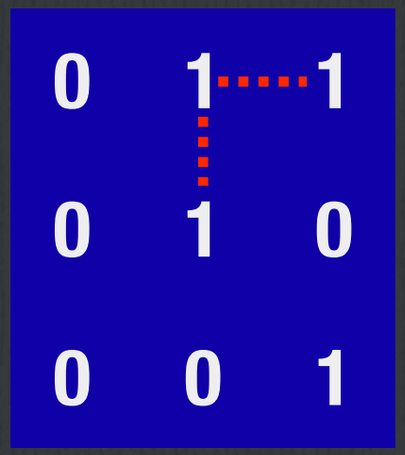
   i) q is in $N_4(p)$, or

   ii) q is in $N_D(p)$ and the set $N_4(p) \cap N_4(q)$ has no pixels whose values are from V

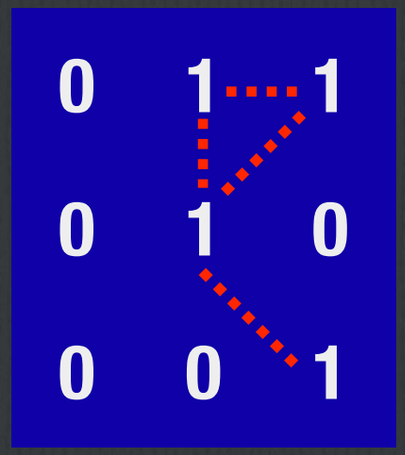
- A (digital) **path** (or curve) from pixel p with coordinates (x,y) to pixel q with coordinates (s,t) is a **sequence of distinct pixels** with coordinates

$$(x_0,y_0), (x_1,y_1),\ldots, (x_n,y_n)$$

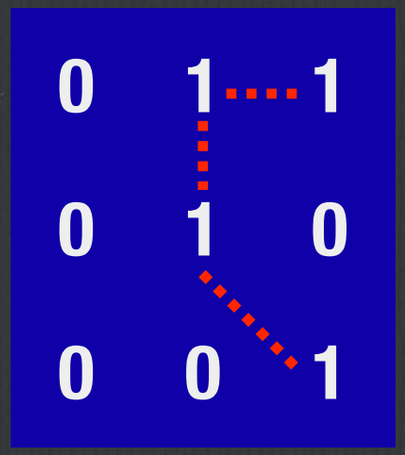
- $(x_0,y_0) = (x,y), \quad (x_n,y_n) = (s,t)$
- Pixels (x_i,y_i) and (x_{i-1},y_{i-1}) are adjacent for $1 \leq i \leq n$
- n is length of the path
- If $(x_0,y_0) = (x_n,y_n)$, the path is a closed path.



4-adjacent

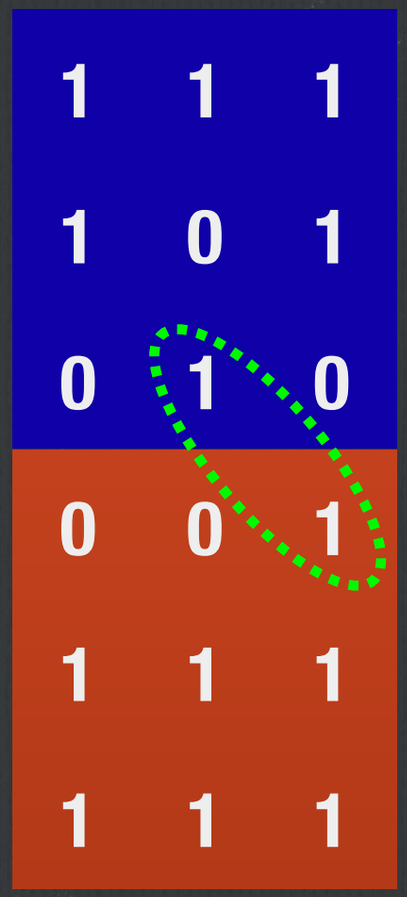


8-adjacent



m-adjacent

- Let S represent a subset of pixels in an image.
  - Pixel p and q are said to be **connected** in S if there exists a path between them consisting entirely of pixels in S.
  - For any pixel p in S, the set of pixels that are connected to it in S is called a **connected component** of S.
  - If it only has one connected component, then S is called a **connected set.**
- Two image area subsets S1 and S2 are connected if some pixel in S1 is adjacent to some pixel in S2.
- Let R be a subset of pixels in an image.
  - We call R a **region** of the image if R is a connected set.
  - Regions R_i and R_j are said to be adjacent if their union forms a connected set.
  - Regions that are not adjacent are said to be **disjoint**.



2 regions (of 1s) that are adjacent if 8-adjacency is used

- The **boundary** (also called the **border** or **contour**) of a region R is the set of points that are adjacent to points in the complement of R.

- The border of a region is the set of pixels in the region that have at least one background neighbor.

0	0	0	0	0
0	1	1	0	0
0	1	1	0	0
0	1	1	1	0
0	1	1	1	0
0	0	0	0	0

The circled point is part of the boundary of the 1-valued pixel only if 8-adjacency between the region and background is used

Distance measures

- For pixels p, q, and z, with coordinates (x,y), (s,t), and (v,w), respectively, D is a distance function or metric if
  - A. $D(p,q) \geq 0$ $(D(p,q) = 0 \text{ iff } p = q)$,
  - B. $D(p,q) = D(q,p)$,
  - C. $D(p,z) \leq D(p,q) + D(q,z)$
- Euclidean distance between p and q is defined as

$$D_e(p,q) = \left[(x-s)^2 + (y-t)^2 \right]^{\frac{1}{2}}$$

  - Radius r centered at (x,y)

- The **city-block distance** between p and q is defined as

$$D_4(p,q) = |x-s| + |y-t|$$

		2		
	2	1	2	
2	1	0	1	2
	2	1	2	
		2		

  - Diamond center at (x,y)
  - $D_4 = 1$ are the 4-neighbors of (x,y)

- The **chessboard distance** between p and q is defined as

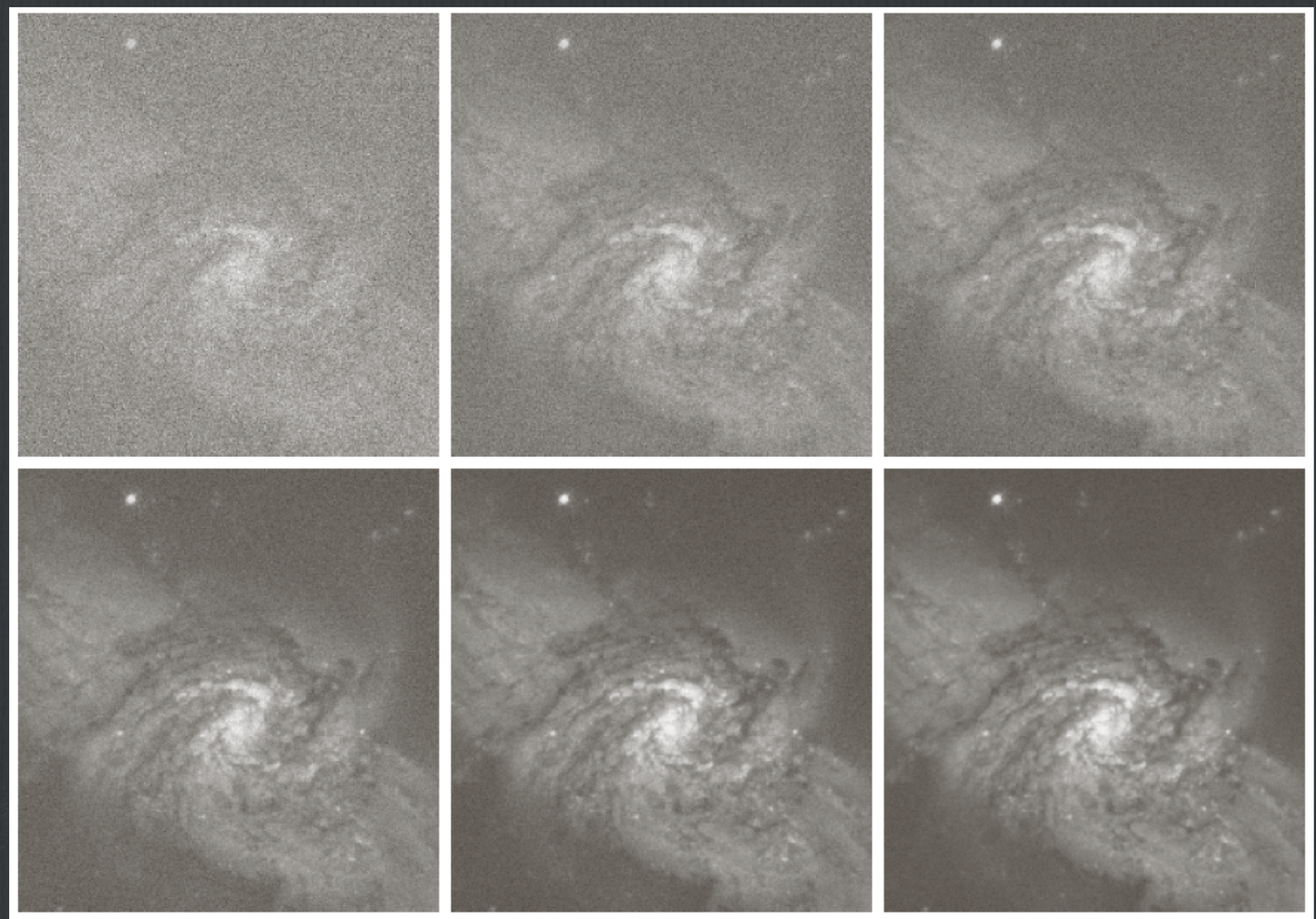
$$D_8(p,q) = \max\left(|x-s|, |y-t|\right)$$

2	2	2	2	2
2	1	1	1	2
2	1	0	1	2
2	1	1	1	2
2	2	2	2	2

  - Square center at (x,y)
  - $D_8 = 1$ are the 8-neighbors of (x,y)

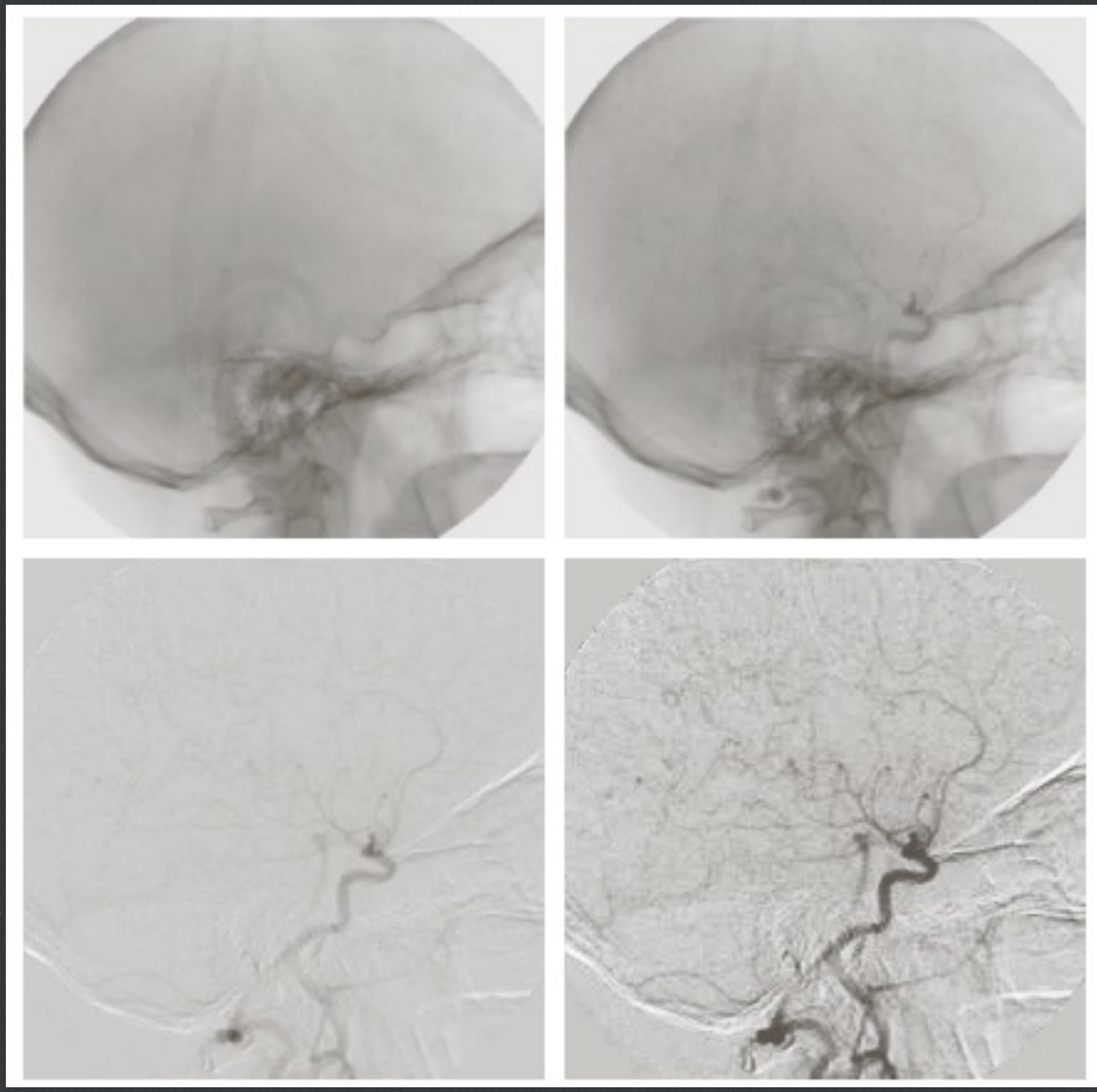
An introduction to the mathematical tools used in digital image processing

- ☐ Array versus matrix operations
- ☐ Linear versus nonlinear operations
- ☐ Arithmetic operations



a b c
d e f

FIGURE 2.26 (a) Image of Galaxy Pair NGC 3314 corrupted by additive Gaussian noise. (b)–(f) Results of averaging 5, 10, 20, 50, and 100 noisy images, respectively. (Original image courtesy of NASA.)



a b
c d

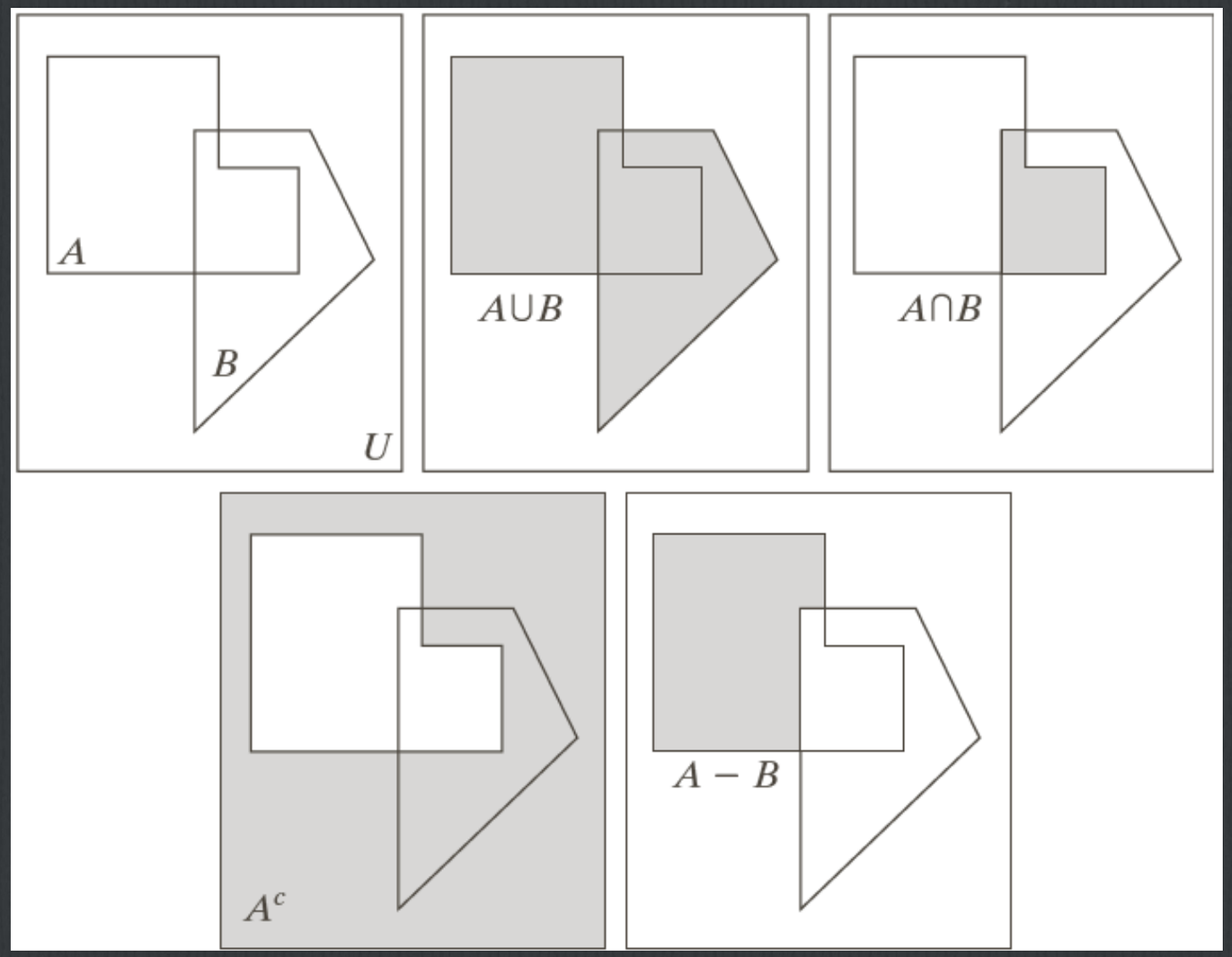
FIGURE 2.28
Digital subtraction angiography. (a) Mask image. (b) A live image. (c) Difference between (a) and (b). (d) Enhanced difference image. (Figures (a) and (b) courtesy of The Image Sciences Institute, University Medical Center, Utrecht, The Netherlands.)



a b c

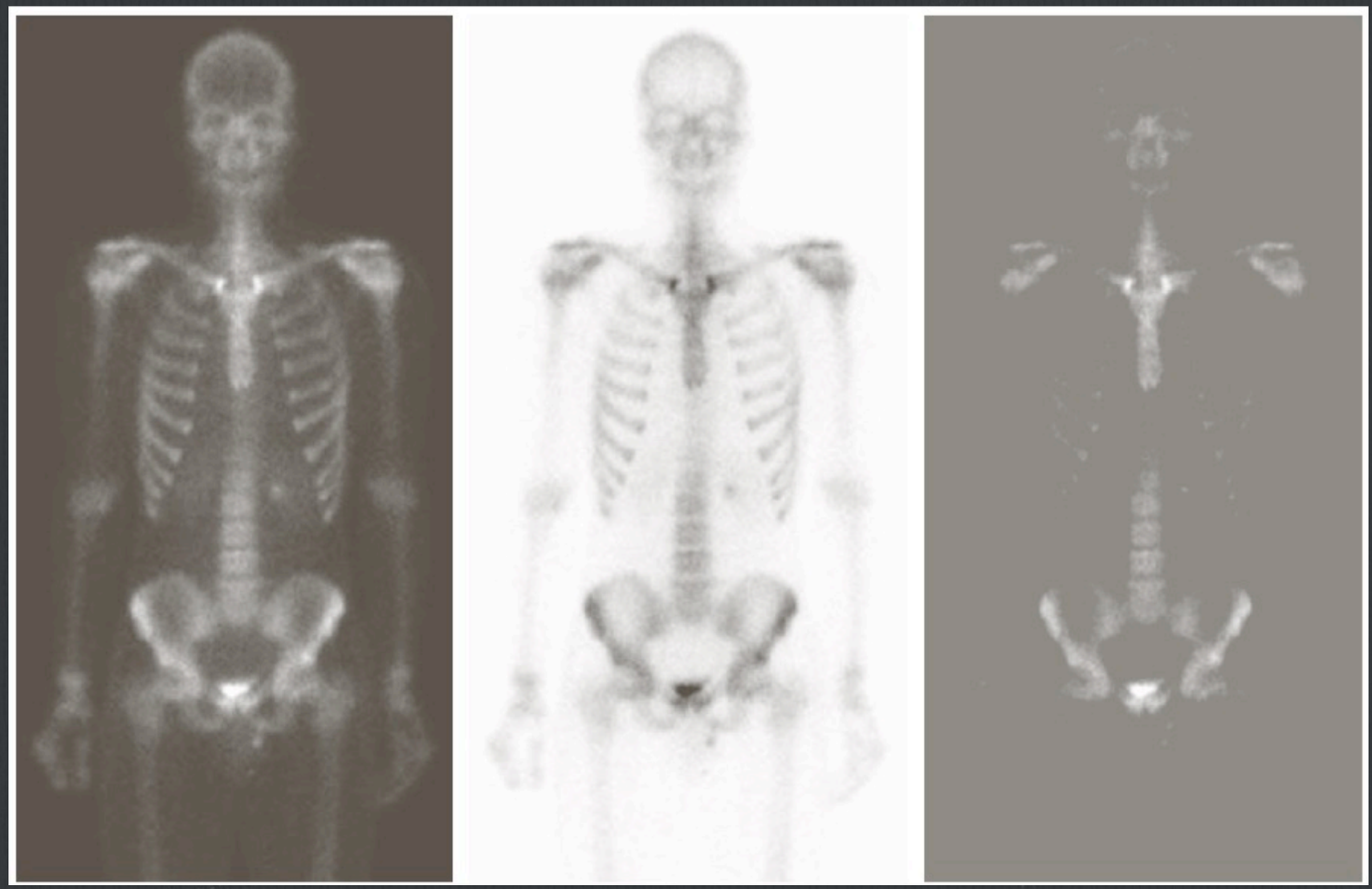
FIGURE 2.30 (a) Digital dental X-ray image. (b) ROI mask for isolating teeth with fillings (white corresponds to 1 and black corresponds to 0). (c) Product of (a) and (b).

Set and logical operations



a b c
d e

FIGURE 2.31
(a) Two sets of coordinates, A and B, in 2-D space. (b) The union of A and B. (c) The intersection of A and B. (d) The complement of A. (e) The difference between A and B. In (b)–(e) the shaded areas represent the member of the set operation indicated.



a b c

FIGURE 2.32 Set operations involving gray-scale images. (a) Original image. (b) Image negative obtained using set complementation. (c) The union of (a) and a constant image. (Original image courtesy of G.E. Medical Systems.)

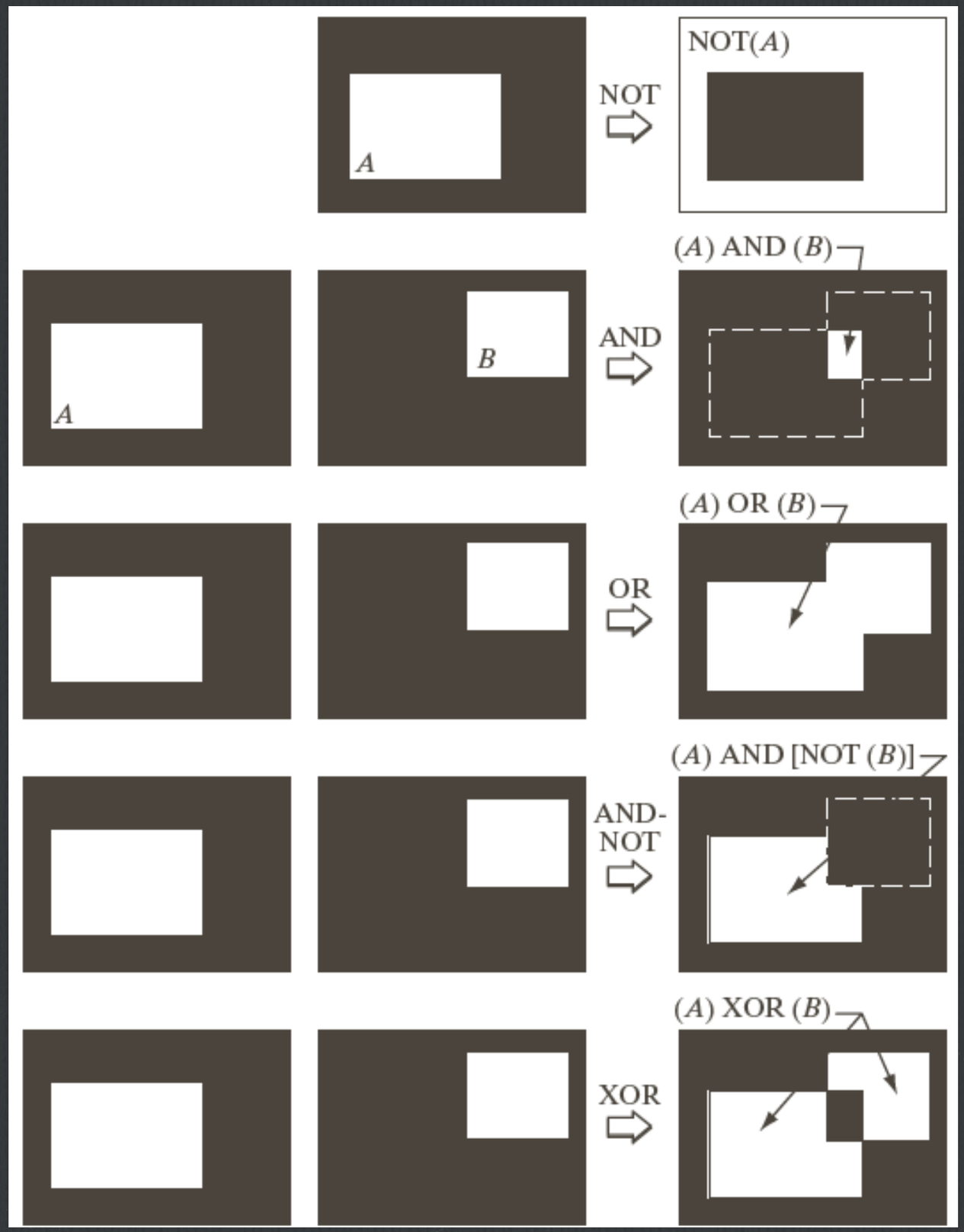
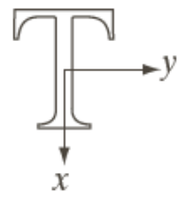
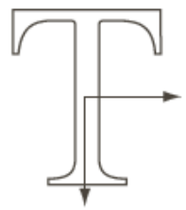
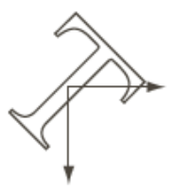
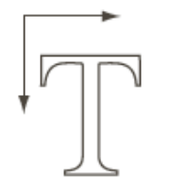
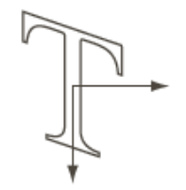


FIGURE 2.33
Illustration of logical operations involving foreground (white) pixels. Black represents binary 0s and white binary 1s. The dashed lines are shown for reference only. They are not part of the result.

Spatial operations

TABLE 2.2
Affine transformations based on Eq. (2.6.–23).

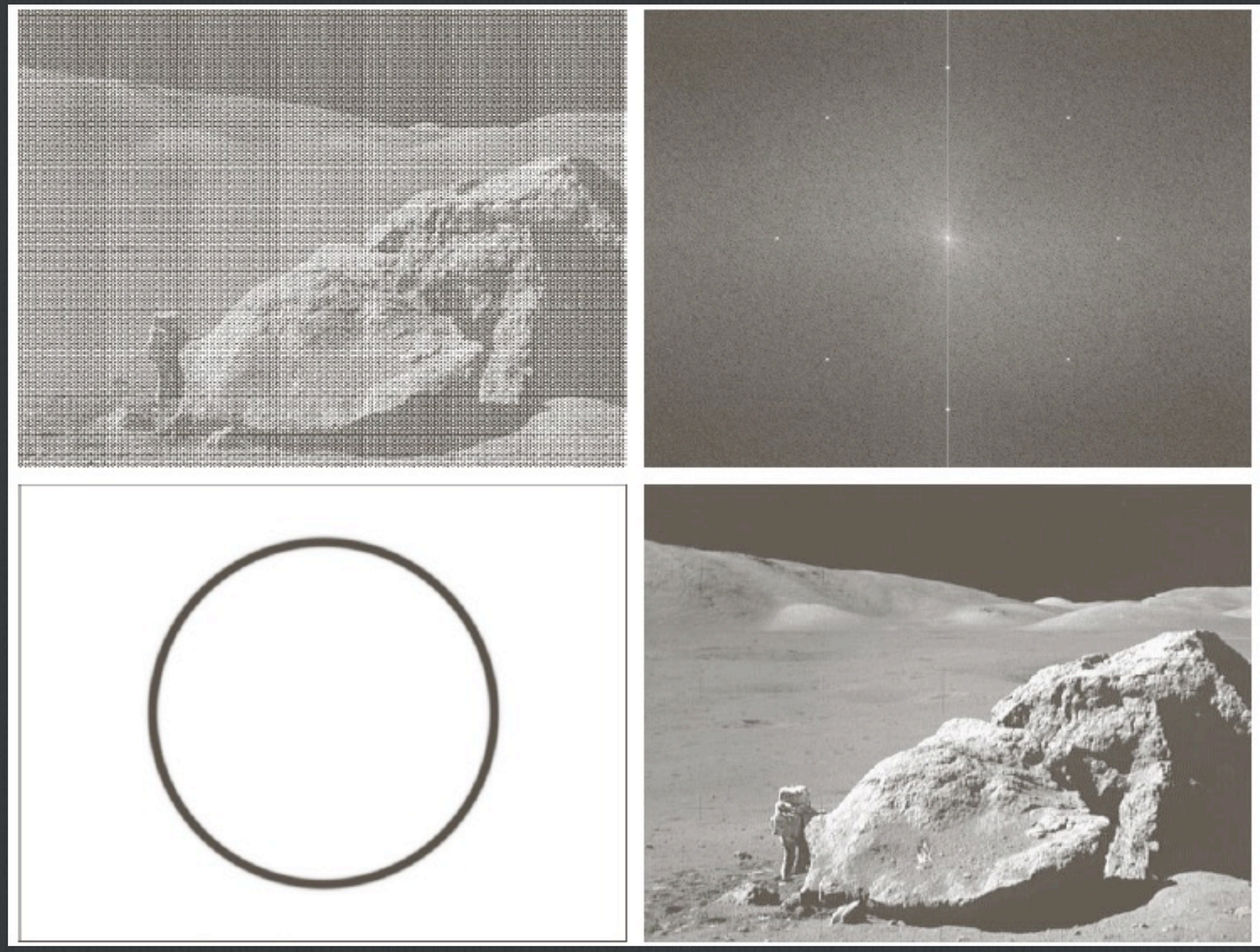
Transformation Name	Affine Matrix, T	Coordinate Equations	Example
Identity	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = w$	
Scaling	$\begin{bmatrix} c_x & 0 & 0 \\ 0 & c_y & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = c_x v$ $y = c_y w$	
Rotation	$\begin{bmatrix} \cos\theta & \sin\theta & 0 \\ -\sin\theta & \cos\theta & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v\cos\theta - w\sin\theta$ $y = v\cos\theta + w\sin\theta$	
Translation	$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ t_x & t_y & 1 \end{bmatrix}$	$x = v + t_x$ $y = w + t_y$	
Shear (vertical)	$\begin{bmatrix} 1 & 0 & 0 \\ s_v & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v + s_v w$ $y = w$	
Shear (horizontal)	$\begin{bmatrix} 1 & s_h & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$	$x = v$ $y = s_h v + w$	



a b c d

FIGURE 2.36 (a) A 300 dpi image of the letter T. (b) Image rotated 21° clockwise using nearest neighbor interpolation to assign intensity values to the spatially transformed pixels. (c) Image rotated 21° using bilinear interpolation. (d) Image rotated 21° using bicubic interpolation. The enlarged sections show edge detail for the three interpolation approaches.

- ☐ Vector and matrix operations
- ☐ Image transforms



a b
c d

FIGURE 2.40
(a) Image corrupted by sinusoidal interference. (b) Magnitude of the Fourier transform showing the bursts of energy responsible for the interference. (c) Mask used to eliminate the energy bursts. (d) Result of computing the inverse of the modified Fourier transform. (Original image courtesy of NASA.)